

Oncology Risk Stratification Analytics

Healthcare | Detailed portfolio case study | Synthetic data

Executive summary

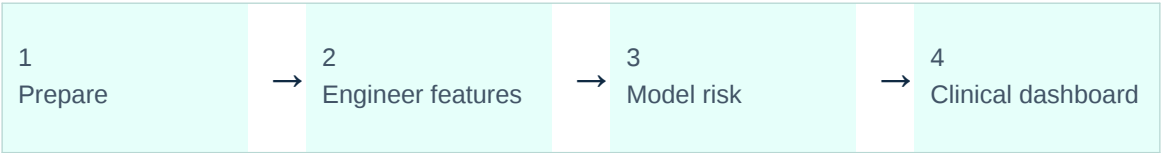
Prepared synthetic clinical data, engineered features, evaluated a risk model and designed decision-support dashboards.

25K patients	18 facilities	87% model accuracy	22% less reporting effort
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Business challenge

Teams needed a consistent view of patient risk, treatment outcomes and facility performance.

Solution architecture



Dashboard mockup

PORTFOLIO DASHBOARD	25K	18	87%	22%
	patients	facilities	model accuracy	less reporting effort
	■■■■■	●●●○	▲ 12%	Status: Review

Illustrative dashboard wireframe. Values are synthetic.

Delivery approach

Inputs	Patient records, Laboratory results, Treatment history, Facility data
Technology	Python, R, Machine Learning, SQL, Tableau
Deliverables	• Synthetic dataset • Feature pipeline • Model prototype • Dashboard concept • Monitoring plan

Key considerations

The design prioritises traceability, clear definitions, appropriate human review, data quality and proportionate monitoring. Any production implementation would require validation against the organisation's data, operating model, policies and applicable requirements.

Portfolio note

This document uses generated data and representative scenarios. It contains no client-confidential information and does not make claims about actual production outcomes.